Foundations Cheat Sheet

Networking, subnetting (binary), coding

BINARY OCTET BIT VALUES

128 64 32 16 8 4 2 1 (192 = 128+64 = 11000000)

CIDR MASKS & BLOCK SIZE (BLOCK = 256 - MASK)

/25 128 · /26 192 · /27 224 · /28 240 · /29 248 · /30 252

USABLE HOSTS (2^HOST - 2)

/24 254 · /25 126 · /26 62 · /27 30 · /28 14 · /29 6 · /30 2

KEY RANGES

Private 10.0.0.0/8, 172.16.0.0/12, 192.168.0.0/16
APIPA 169.254.0.0/16 · Loopback 127.0.0.1 · IPv6 loopback ::1
Classes A 1-126 · B 128-191 · C 192-223

PORTS

20/21 FTP · 22 SSH · 23 Telnet · 25 SMTP · 53 DNS · 67/68 DHCP · 80 HTTP · 88 Kerberos
110 POP3 · 123 NTP · 143 IMAP · 161 SNMP · 389 LDAP · 443 HTTPS · 445 SMB · 3389 RDP

MODELS & PROTOCOLS

OSI Physical, Data Link, Network, Transport, Session, Presentation, Application
TCP handshake SYN, SYN-ACK, ACK · TCP reliable / UDP fast
Helpers DNS names->IP · DHCP DORA · NAT private<->public · ARP IP->MAC

PYTHON QUICK REFERENCE

def f(x): ... for x in list: if/elif/else try/except
format(n,'08b') len(list) list [] / dict {} / set() / tuple ()
Big-O index O(1) · binary search O(log n) · linear O(n) · nested O(n^2)

Security+ Cheat Sheet

SY0-701 · 90 min, up to 90 Qs, pass 750/900, valid 3 yrs

FOUNDATIONS

CIA Confidentiality, Integrity, Availability · AAA = Authn, Authz, Accounting
Data states At rest, in transit, in use

CRYPTOGRAPHY

Symmetric One shared key, fast: AES, ChaCha20, 3DES
Asymmetric Key pair: RSA, ECC, Diffie-Hellman (key exchange)
Hashing One-way integrity: SHA-256 (MD5/SHA-1 weak) · salt + key stretching (bcrypt, PBKDF2)
PKI CA issues certs · CRL/OCSP revocation · CSR request · TLS secures HTTPS

CONTROLS

Function Preventive, detective, corrective, deterrent, compensating, directive
Category Technical, managerial, operational, physical

ATTACKS

Phishing / smishing / vishing / whaling · on-path (MITM) · SQLi · XSS · privilege escalation
DDoS · replay · pass-the-hash · supply chain · watering hole · zero-day

IDENTITY & NETWORK DEFENSE

MFA factors Know, have, are, somewhere, do · TOTP, FIDO2/passkey
Tools SIEM, SOAR, EDR, DLP · IDS detect / IPS block · WAF, NGFW, DMZ, 802.1X

OPERATIONS & GOVERNANCE

IR lifecycle Prepare, detect, analyze, contain, eradicate, recover, lessons learned
Resilience Backups: full/incremental/differential · RTO (downtime) vs RPO (data loss)
Governance Risk = likelihood x impact · policies, third-party risk, compliance (GDPR)

RHCSA Cheat Sheet

EX200 (RHEL 10) · 2.5 hrs, hands-on, pass 210/300

USERS, GROUPS, PERMISSIONS

useradd -u 1500 -G devs bob   passwd bob   usermod -L bob
chmod 640 f   chown user:grp f   setfacl -m g:devs:rwx /data   getfacl /data

Numeric r=4 w=2 x=1 · 755=rwx r-x r-x · 640=rw- r-- --- · umask 022

Special bits SUID 4xxx (as owner) · SGID 2xxx (group) · sticky 1xxx (/tmp)

STORAGE & LVM

lvcreate -L 1G -n lvdata vgdata   mkfs.xfs /dev/vgdata/lvdata
echo '/dev/vgdata/lvdata /data xfs defaults 0 0' >> /etc/fstab   mount -a

Grow lvextend -L +1G … then xfs_growfs /data · lsblk, df -h

SERVICES, BOOT, CRON

systemctl start|enable|status|reload|mask <svc>   journalctl -u <svc>

Targets systemctl set-default multi-user.target (CLI) / graphical.target · crontab -e

NETWORK & FIREWALL

nmcli …   firewall-cmd --permanent --add-service=http   firewall-cmd --reload
ssh-keygen   ssh-copy-id user@host   ss -tulpn

SELINUX

getenforce   setenforce 0   restorecon -Rv /path   setsebool -P <bool> on

Ports/contexts semanage port -a -t <type> -p tcp <port> · semanage fcontext …

CONTAINERS & PACKAGES

podman run …   podman images   dnf install <pkg>   dnf list installed

AWS Cheat Sheet

CCP (CLF-C02, pass 700) then SAA (SAA-C03, pass 720)

COMPUTE & STORAGE

Compute EC2 (VMs) · Lambda (serverless, 15-min max) · ECS/EKS/Fargate (containers)
Storage S3 (object, 11 nines) · EBS (block) · EFS (shared file) · Glacier (archive)
S3 Classes Standard/IA/One Zone/Glacier/Intelligent · versioning, lifecycle, SSE-KMS

DATABASE

Relational RDS / Aurora · Multi-AZ = failover, read replica = read scaling
Other DynamoDB (NoSQL) · ElastiCache (cache) · Redshift (warehouse) · Athena (SQL on S3)

NETWORKING & SECURITY

VPC Public subnet -> Internet Gateway · Private -> NAT Gateway · endpoints, peering, TGW
Firewalls Security Group = stateful, instance · NACL = stateless, subnet
IAM Users (long-term) vs Roles (temporary, preferred) · JSON policies · least privilege · global
Keys/edge KMS (keys) · CloudFront (CDN) · Route 53 (DNS) · WAF/Shield

INTEGRATION & OPS

Decouple SQS (queue) · SNS (pub/sub) · EventBridge · Step Functions
Observe CloudWatch (metrics/logs) · CloudTrail (API audit) · Trusted Advisor

EXAM ESSENTIALS

Pricing On-demand · Reserved/Savings Plans · Spot (interruptible, cheapest)
Pillars Ops Excellence, Security, Reliability, Performance, Cost, Sustainability
Shared resp. AWS secures the cloud; you secure what is in it (data, config, patching EC2 OS)

PMP Cheat Sheet

2026 ECO · domains + agile; needs 35 hrs + experience

STRUCTURE

Domains People ~42% · Process ~50% · Business Environment ~8% (about half agile/hybrid)
Process groups Initiating, Planning, Executing, Monitoring & Controlling, Closing

THE PMI MINDSET (ANSWERS MOST QUESTIONS)

Be a proactive servant leader · understand the root cause first · empower the team · collaborate

EARNED VALUE & FORMULAS

EVM SPI = EV/PV (schedule) · CPI = EV/AC (cost) · below 1 = behind/over
Estimate PERT = (O + 4M + P)/6 · Channels = n(n-1)/2
Reserves Contingency = known risks (in baseline) · Management = unknown (outside)

SCHEDULE & SCOPE

Scope WBS -> work packages · scope creep (uncontrolled) · gold plating (unrequested)
Schedule Critical path = longest · float = slack · crashing (add resources) vs fast tracking (overlap)
Dependencies FS, SS, FF, SF · lead vs lag

AGILE

Scrum Roles: PO, Scrum Master, Developers · Events: Sprint, Daily, Review, Retro
Artifacts Product Backlog, Sprint Backlog, Increment · velocity · Definition of Done

CONTRACTS & PEOPLE

Contracts Fixed price (seller risk) · cost-plus (buyer risk) · T&M
Teams Tuckman: forming, storming, norming, performing, adjourning · Theory X/Y

CISSP Cheat Sheet

8 domains · think like a risk manager

DOMAINS (8)

1 Security & Risk Mgmt · 2 Asset Security · 3 Architecture & Engineering · 4 Comm & Network
5 IAM · 6 Assessment & Testing · 7 Security Operations · 8 Software Development Security

RISK

Formulas Risk = Threat x Vulnerability · ALE = SLE x ARO · SLE = Asset Value x Exposure Factor
Responses Avoid, transfer (insurance), mitigate, accept · residual risk remains after controls

ACCESS CONTROL

Models DAC (owner) · MAC (labels) · RBAC (role) · ABAC (attributes)
Principles Least privilege, need to know, separation of duties, defense in depth

SECURITY & THREAT MODELS

Confidentiality Bell-LaPadula: no read up, no write down
Integrity Biba: no write up, no read down · Clark-Wilson: well-formed transactions
STRIDE Spoofing, Tampering, Repudiation, Info disclosure, DoS, Elevation

CRYPTO

Keys Symmetric n(n-1)/2 · use GCM not ECB · Kerberos (KDC/TGT) · certs signed by a CA

OPERATIONS & CONTINUITY

IR Prepare, detect, respond, mitigate, report, recover, remediate, lessons
BCP/DRP BIA -> MTD · RTO < MTD · RPO (data loss) · hot/warm/cold sites
Frameworks NIST CSF (Identify, Protect, Detect, Respond, Recover) · ISO 27001 · ethics: society first

AI Innovator Cheat Sheet

Building with Claude, Codex, and agents

CORE TERMS

Model LLM predicts tokens (~3/4 word) · context window = max tokens · temperature = randomness
Adaptation Prompting (instructions) · RAG (retrieved context) · fine-tuning (weights: SFT/DPO/RFT)

RAG

Pipeline Chunk -> embed -> store (vector DB) -> retrieve top-k -> stuff prompt -> generate
Vector DBs Pinecone, Qdrant, Weaviate · semantic search via cosine similarity · reranking
RAG Triad Faithfulness, Answer Relevance, Context Relevance (measure with RAGAS)

AGENTS

Definition LLM + tools + loop + memory · loop: perceive, plan, act, observe, repeat (ReAct)
MCP Model Context Protocol: standard to connect agents to tools/data · Agent Skills = SKILL.md
Orchestrate LangGraph, CrewAI · guardrails/hooks · human-in-the-loop

PROMPT ANATOMY

Role + task + examples + output format + delimited input (few-shot, chain-of-thought)

PRODUCTION / LLMOPS

Ship FastAPI + Docker on AWS · observability: Langfuse, LangSmith · sandbox: E2B, Modal
Evals Track accuracy, latency, cost on every change · LLM-as-judge · defend prompt injection

TOOLS & PORTFOLIO

Tools Claude Agent SDK (pip install claude-agent-sdk) · OpenAI Codex (GPT-5.5)
Ship 3-5 RAG assistant · FastAPI+Docker LLM API · multi-agent workflow · MCP assistant · instrumented

Target Roles & Priority Map

AI Business Architect / Evangelist + Forward Deployed Engineer

TWO ADOBE-STYLE TARGET ROLES (A SPECTRUM)

Architect / Evangelist Business architecture + AI adoption; light coding; BI dashboards; TOGAF, PL-300, AI-900, Prosci
Forward Deployed Eng. Software engineering + MLOps; strong Python/PyTorch; SQL + evals + observability; AWS + Docker + CI/CD
Shared core AI literacy, RAG, agents, evals, customer empathy, communication

A2A MEANS TWO DIFFERENT THINGS

Architect role A2A = Agent-to-Agent (Google protocol: agents coordinate)
FDE role A2A = application-to-application (REST/gRPC, webhooks, events)

REPRIORITIZED ORDER (ROLE-FIRST)

1 Applied AI & agents · 2 Enterprise AI tools + protocols (AI-900) · 3 Business architecture (TOGAF)
4 Data & BI (PL-300) · 5 AI adoption (ADKAR/Prosci) · 6 CAPM/PMP · 7 AWS CCP · 8 Python
Engineer lean adds Track F (MLOps) + AWS SAA, Docker, CI/CD. RHCSA/CISSP optional unless a role needs them.

PORTFOLIO BEATS CERTS (BUILD ALL OF THESE)

MCP agent · Copilot Studio chatbot · capability map + value stream · Power BI adoption dashboard
AI enablement plan · an AI-backed API with tests, an eval harness, tracing, and a dashboard

Business Architecture Cheat Sheet

TOGAF Standard 10th Edition · BIZBOK

CORE IDEA

Business architecture links strategy to capabilities, value streams, and processes: the what before the how.

CAPABILITY MAP

Definition A stable, structure-agnostic view of what the business does (not how)
Dimensions People, Process, Information, Resources
Why stable Ignores org charts, which change often

VALUE STREAM

Definition End-to-end stages that deliver value to a stakeholder
Mapping Map capabilities to the stages that need them; heat-map for investment

TOGAF ADM

Method Phases A-H; Phase B = Business Architecture
Flow Current-state assessment -> target architecture -> gap analysis -> roadmap

MODELING & TOOLS

Diagrams ArchiMate (architecture) · BPMN (processes) · draw.io / Archi (free)
Body of knowledge BIZBOK for capability and value-stream practice

CERT

Target TOGAF Enterprise Architecture Foundation, then Practitioner
Prove it A capability map + a value stream for a real or sample business

Data Analytics & BI Cheat Sheet

Power BI (PL-300), Tableau, Excel

THE BI PIPELINE

Extract/Transform/Load (Power Query) -> model (star schema) -> measures (DAX) -> visualize -> story

MODELING

Star schema Fact table (events/measures) linked to dimension tables (attributes)
Measure A calculated aggregate, e.g. total sales
DAX context Row context (current row) vs filter context (applied filters)

VISUALS

Trend over time Line chart
Comparison Bar chart · Parts of a whole: pie (use sparingly)
Interactivity Slicers filter the report; a dashboard = one view of key KPIs

EXCEL & TABLEAU

Excel PivotTables + Power Query for cleaning and aggregation
Tableau Alternative viz tool; Desktop Specialist cert optional

ADOPTION METRICS (FOR THE EVANGELIST ROLE)

Track Active users, usage rate, time saved, ROI
Rule Prefer actionable metrics over vanity metrics

CERT

Target Microsoft PL-300 (Power BI Data Analyst)
Prove it An AI-adoption dashboard built in Power BI

AI Adoption & Change Cheat Sheet

Evangelism, ADKAR, metrics

ADKAR (PEOPLE ADOPT IN THIS ORDER)

A Awareness of the need to change
D Desire to support the change
K Knowledge of how to change
A Ability to apply the new skills
R Reinforcement to sustain it
Diagnose a stuck rollout by finding the first missing letter.

RUN ADOPTION LIKE A PRODUCT

Steps Pilot to prove value -> recruit AI champions -> workshops -> Center of Excellence
Resistance Address the why, not just the how

MEASURE & REPORT

Metrics Adoption rate, monthly active users, time saved, ROI
Exec reporting Impact, progress, and the next ask, kept simple
Lead vs lag Leading predicts; lagging confirms

EVANGELIZE

Build in public Publish what you build; present to varied audiences
Audience range VP-level to individual contributor

CREDENTIAL

Valued Prosci Change Management Certification (ADKAR is free to study)
Prove it An AI enablement plan + a workshop outline + an adoption dashboard

Enterprise AI Tools & Agent Protocols

Copilot, Glean, Cursor, Replit · MCP, A2A

ENTERPRISE AI TOOLS

Copilot / Copilot Studio M365 assistant; Copilot Studio builds low-code grounded agents
Glean Enterprise AI search/assistant over company data
Cursor / Replit AI code editor; browser IDE with an AI build agent
Adobe Firefly (images), Express, GenStudio (content supply chain)

AGENT PROTOCOLS (THE 2026 STACK)

MCP Model Context Protocol (Anthropic): agent -> tools and data
A2A Agent-to-Agent (Google): agents discover and delegate
Two-layer stack MCP vertical (tools) + A2A horizontal (agents)
Also ACP/UCP (agent commerce); OAP in the open-protocol landscape

DATA STRUCTURES FOR AI

Vector database Embeddings for similarity search (Pinecone, Qdrant, Weaviate)
Knowledge graph Entities + relationships (Neo4j); reasoning over connected data

CERT & PROOF

Cert Microsoft AI-900 (Azure AI Fundamentals)
Prove it A grounded Copilot Studio chatbot + an MCP-connected agent

AI Engineering & MLOps Cheat Sheet

Forward Deployed Engineer path

BUILD

Language / frameworks Python + PyTorch (tensors, training loop); FastAPI for services
Dev tooling Git, CI/CD, Docker; tests on everything

INTEGRATION (APPLICATION-TO-APPLICATION)

APIs REST (JSON/HTTP) and gRPC (binary/HTTP2)
Async Webhooks, queues, event-driven systems
Reliability Auth, rate limits, retries with backoff, idempotency

EVALUATION

Offline Golden test sets + acceptance thresholds
Online A/B tests, user rating flows, guardrails
Data work SQL + notebooks; segment metrics by customer/cohort/use case

OBSERVABILITY & LLMOPS

Three pillars Logs, metrics, traces (OpenTelemetry)
Stack Prometheus + Grafana, Datadog; Langfuse/LangSmith for LLM traces
Safe rollout Feature flags + canary releases; watch for model drift
LLMOps Prompt/version management, eval harnesses, cost tracking

PROOF

Portfolio An AI-backed API with tests, an eval harness, tracing, and a dashboard

Role Decision, 90-Day Start & Degrees

Pick a lean, start fast, and decide on a degree

WHICH LEAN?

Forward Deployed Eng. You love building/debugging code, data pipelines, evals, observability; customer-facing engineer
Architect / Evangelist You love translating needs, workshops, dashboards, change management, presenting to leaders
Unsure Do the shared core (AI literacy, RAG, agents, evals, communication), decide at day 90

90-DAY START — ARCHITECT & EVANGELIST

Wk 1-4: prompt eng + a Copilot Studio chatbot; start AI-900
Wk 5-8: TOGAF basics; capability map + value stream; begin Power BI
Wk 9-12: Power BI adoption dashboard; AI enablement plan (ADKAR); present; sit AI-900

90-DAY START — FORWARD DEPLOYED ENGINEER

Wk 1-4: Python + Git/tests/Docker; a small AI-backed FastAPI service
Wk 5-8: RAG + an MCP agent; PyTorch basics; add REST/gRPC + webhooks
Wk 9-12: eval harness (golden set, A/B); add tracing + a metrics dashboard
Either lean: ship ONE working, instrumented portfolio piece by day 90.

DEGREES (WORTH IT?)

Best ROI MS Computer Science AI/ML online: Georgia Tech OMSCS (~$9k) or UT Austin (~$10k), no GRE
Architect lean MS Information Systems or MBA (Adobe architect role prefers an MBA)
Middle MS Data Science or AI (~$10-20k)
Doctorate PhD only for AI research scientist roles; not needed here. DBA is a much-later option
Timing Start a master's AFTER shipping a portfolio piece; portfolio + certs still lead

Communication Cheat Sheet

Mumbling to paid speaker, in five rungs

FIX MUMBLING (RUNG 1)

Breathe Diaphragmatic (belly) breathing for steady, supported volume
Slow down ~130-150 words/min; open your mouth more; over-enunciate
Pause Silence adds clarity and authority; replace um/uh with a pause
Feedback Record 1 minute daily and review; a voice coach or SLP speeds it up

CONVERSATION (RUNG 2)

System Prepared openers + active listening + go-to questions
Opener Ask what someone is working on or enjoying

PRESENTATION (RUNG 3)

Structure Hook, three points, close with a callback to the hook
Story Situation, complication, resolution, lesson
Slides One idea per slide; visuals over text; signpost each part

STAGE (RUNG 4)

Practice Toastmasters + local meetups = graded, low-stakes reps
Nerves Ground and breathe first; reframe adrenaline as energy

PROFESSIONAL (RUNG 5)

Platform Signature topic (your AI/cyber expertise) + demo reel + one-sheet
Path Meetups -> conferences -> paid gigs; build in public along the way

WORK WITH YOUR WIRING

Scripts Prepare and rehearse to automaticity; running on a script is a strength
Sensory Manage lights/noise; visit the room first; water nearby; recover after
Kindness A stumble is a rep, not a verdict; regulate before you speak

GRC, NIST, Auditing & IAM

Cybersecurity depth around CISSP

NIST FRAMEWORKS

CSF 2.0 Govern, Identify, Protect, Detect, Respond, Recover
RMF (800-37) Categorize, Select, Implement, Assess, Authorize, Monitor
Controls SP 800-53 (control catalog) · SP 800-171 (CUI)

AUDITING & COMPLIANCE

Standards ISO 27001 · SOC 1/2 (Type I design, Type II effectiveness) · PCI DSS · HIPAA · GDPR
Basics Control vs evidence · gap assessment · three lines of defense
Risk Inherent vs residual · risk register · compliance is not security
Certs CISA (audit) · CRISC (risk)

IDENTITY & ACCESS MANAGEMENT

Core AuthN (who) vs AuthZ (what you may do) · least privilege
SSO/Federation SAML, OAuth 2.0, OIDC · MFA · directories (AD, Entra ID)
Access models RBAC (by role) vs ABAC (by attributes)
Lifecycle & govern Joiner-Mover-Leaver · IGA · PAM (privileged accounts)
Zero Trust Never trust, always verify; identity is the new perimeter
Cert Microsoft SC-300 (Identity and Access Administrator)

PROVE IT

Portfolio A CSF-mapped risk assessment + a simple IAM design (roles, MFA, least privilege)

Your Mind & Your Message

Cognitive manual + how to communicate and interview

HOW YOUR MIND WORKS

You are Top-down, pattern-recognizing, abstract, non-linear, meaning-driven
Strengths Systems thinking, strategy, teaching, deep focus on meaningful goals
Friction Fine-motor/linear/rote tasks; needing the why; social/sensory load; change

HOW TO FEED YOU INFORMATION

1 Big picture and why first · 2 Answer first (BLUF), then support · 3 Tie it to the long-term purpose
4 Patterns/analogies/diagrams over step lists · 5 Steps only when asked · 6 Direct; one next action

EXECUTIVE COMMUNICATION (YOUR SUPERPOWER)

Lead with the point Answer or insight first, then why, then detail on demand
Reframe The why-first mind that got a C in rote class is what executives want

INTERVIEW PLAYBOOK

Tell me about yourself 60-90s purpose arc: start, the pattern, where you aim and why (scripted)
Answers Result-first STAR; 5-6 rehearsed stories, each proving a strength
Nerves & senses 3 breaths first; reduce glare/noise; water; arrive early
Small talk 3 openers ready; it is a warm-up, not a test
Interview them back Screen for autonomy and explained decisions; you need why-driven managers

JOB SEARCH

Positioning I see the system and the why fast; I connect AI, security, and business
Approach Build in public; warm intros; batch interviews; recover after

Founder Path & Daily Operating System

2-3 year plan + daily and decision frameworks

THE THROUGH-LINE

Your four missions (teach, independence, protect, frontier AI) converge on one thing:
help people adopt AI safely. That is a job now (AI enablement / evangelist) and a business later.

RECOMMENDED VENTURE

Model AI + security enablement: productized consulting and education, then leverage (course/community/tool)
Engine Speaking + building in public builds the audience and authority (2026: audience beats product)

STAIR-STEP TO FULL-TIME FOUNDER

Year 1 Authority & audience: ship projects, post weekly, start speaking, small email list, 1-2 advisory gigs
Year 2 Validated income: a paid workshop/cohort/retainer; first clients; guest on podcasts and stages
Year 3 Transition: scale + add leverage; build 6-12 months runway; go full-time when the pipeline justifies
De-risk Keep the job as runway and R&D; validate with paying clients before leaving

DAILY OPERATING SYSTEM

Morning 3 breaths, read your why, pick 1-3 top-down priorities; hardest/most meaningful first
Day Protect flow blocks; batch social/admin; timer to start
Evening 2-min shutdown: what moved, one win, tomorrow's first task; then stop
Weekly 15-min review: progress vs goal, one adjustment, celebrate one finished thing

DECISION FRAMEWORKS

Why filter Does it serve the long-term purpose? If not: drop, shrink, or delegate
Reversible? Move fast on reversible; go slow only on the few irreversible ones
Time-box Decide within a deadline at good-enough; refine later
Delegate low-level Outsource fine-motor/linear/admin; let AI write the steps you skip

Supply Chain & Cost Modeling

The transferable core skill of procurement

COST MODELING: THREE FRAMEWORKS

Should-cost Materials + Labor + Overhead + Other + Margin (your bottom-up estimate)
TCO Purchase + Maintenance + Operating + Logistics + Risk + Disposal (whole life)
Parametric / ABC Cost from a driver (per lb) / allocate overhead by activity used

WORKED EXAMPLE (SHOULD-COST)

$55 material + $30 labor + $25 overhead = $110 base; +15% margin ($16.50) = $126.50
A $150 quote becomes a specific debate about the extra $23.50, not a vague discount

NEGOTIATION LEVERS (MOST TO LEAST)

Order Margin -> labor & overhead -> logistics -> materials (materials usually market-driven)
Shift Evidence-based (cost build-up) beats pressure-based (budget target)

THE CONNECTED FUNCTION

Cost links all Sourcing (total cost) -> contracts (price mechanisms) -> supplier performance
SCOR map Plan, Source, Make, Deliver, Return, Enable
Key terms CLM, category management, spend analysis, three-way match, S&OP, safety stock, bullwhip

DEFENSE / AEROSPACE CONTEXT

Compliance FAR / DFARS; ITAR / EAR export controls; approved suppliers
Tools & terms SAP Ariba (source-to-pay); cost avoidance vs cost savings; Lean supply

STUDY & CERTS

Videos Should-Cost Analysis (TCO vs Price); Procurement Tactics; CIPS; MITx CTL
Certs CPSM (ISM) - US gold standard; CSCP & CPIM (ASCM); CIPS (global)
Your angle East West is a contract manufacturer; you ran the ERP where cost and sourcing live

Methods, PM & Cross-Functional Fluency

Enough to speak with confidence, then depth where needed

AGILE & SCRUM

Roles / events PO, Scrum Master, Developers / Sprint, Daily, Review, Retro
Cert PSM I (Scrum.org); free Open Assessments to practice

LEAN SIX SIGMA

Lean / Six Sigma Eliminate waste (8 wastes: DOWNTIME) / reduce variation
DMAIC Define, Measure, Analyze, Improve, Control; Yellow/Green Belt

SYSTEMS ENGINEERING & PM

Frame Requirements, V-model, MBSE; INCOSE, CSEP later
Design-to-cost Cost as a design constraint; pair cost + physical models (TPM edge)

E-PROCUREMENT

Platforms SAP Ariba, Coupa, Jaggaer (source-to-pay)
Flow P2P; RFx (RFI/RFP/RFQ); e-auctions; punch-out catalogs

CONVERSATIONAL TECH FLUENCY

Python / PowerShell General scripting & data / Windows automation
REST API HTTP methods (GET/POST/PUT/DELETE) + JSON; endpoints, webhooks, tokens
Anchor You did ETL + SQL at East West and Aptean; speak from that

DATA, FINANCE, SALES

Data Pivot tables, Power BI; descriptive vs predictive
Finance P&L, ROI, CapEx vs OpEx, business case
Sales Consultative selling, discovery, customer success (retention & expansion)

SQL & Excel Cheat Sheet

The two most cross-role practical skills

SQL CLAUSE ORDER

SELECT ... FROM ... WHERE ... GROUP BY ... HAVING ... ORDER BY ... LIMIT

CORE SQL

```
SELECT status, COUNT(*), SUM(total) FROM orders GROUP BY status;
SELECT c.name, o.total FROM customers c JOIN orders o ON o.customer_id=c.id;
-- no-match: LEFT JOIN orders o ON ... WHERE o.id IS NULL
```

WHERE vs HAVING Rows before grouping vs groups after

Careful UPDATE/DELETE always need a WHERE, or you hit every row

Practice sqliteonline.com, SQLBolt, Mode SQL Tutorial

EXCEL SHORTCUTS

Navigate Ctrl+Arrow (edge) - Ctrl+Shift+Arrow (select to edge) - Ctrl+End

Build Alt+= AutoSum - F4 absolute ($) - Ctrl+T Table - Ctrl+Shift+L filter

Edit F2 edit - Ctrl+Enter fill selection - Ctrl+1 format cells

EXCEL FUNCTIONS

Lookup XLOOKUP (over VLOOKUP), INDEX+MATCH

Conditional SUMIFS, COUNTIFS, IF, IFERROR

Text/dynamic TRIM, CONCAT/&, UNIQUE, FILTER, SORT

Power tools PivotTables and Power Query (Get & Transform)

Azure Fundamentals (AZ-900) + AWS Map

Cloud recall for interviews

AZURE CORE SERVICES

Compute VMs (IaaS), App Service (PaaS), Functions (serverless), AKS (containers)
Storage Blob Storage (objects), Files, Disks
Databases Azure SQL (relational), Cosmos DB (NoSQL), Synapse (analytics)
Network Virtual Network (VNet), Load Balancer, VPN Gateway
Identity Microsoft Entra ID (was Azure AD) + RBAC
Manage Resource Groups, ARM, Azure Monitor, Cost Management, Policy
AI Azure AI services, Azure OpenAI, Azure ML
Hierarchy Management group > subscription > resource group > resource

AZURE TO AWS QUICK MAP

VM / storage VM = EC2 ; Blob = S3
Serverless / SQL Functions = Lambda ; Azure SQL = RDS
NoSQL / network Cosmos DB = DynamoDB ; VNet = VPC
Identity / monitor Entra ID + RBAC = IAM ; Azure Monitor = CloudWatch

CERTS

Azure AZ-900 then AZ-104; AI-900, PL-300 pair well
AWS CLF-C02 (Cloud Practitioner) then SAA-C03

Binary & Subnetting Deep Dive

Count, convert, subnet, fast

COUNTING IN BINARY

Bit values 128 64 32 16 8 4 2 1 (each doubles, right to left)

0000=0 0001=1 0010=2 0011=3 0100=4 0101=5 0110=6 0111=7 1000=8 … 1111=15

Powers of 2 1 2 4 8 16 32 64 128 256 512 1024

CONVERT

Dec -> bin Subtract largest bit that fits. 200 = 128+64+8 = 11001000

Bin -> dec Add the 1-position values. 10101100 = 172

Hex 0-9 then A-F; one hex digit = 4 bits; FF = 11111111 = 255

SUBNETTING

Masks /25=128 /26=192 /27=224 /28=240 /29=248 /30=252

Formulas Usable hosts = 2^host - 2 ; Block size = 256 - mask

Fast method Interesting octet -> block size -> list multiples -> round down; broadcast = next net - 1

Worked /26 192.168.1.0/26: block 64 -> .0/.64/.128/.192; .0 net, .63 bcast, .1-.62 hosts

VLSM Allocate the largest subnet first

Wildcard mask Inverse of the subnet mask; /24 -> 0.0.0.255

DRILL (ANSWERS)

/27 usable = 30 ; .100 in /26 = the .64 subnet ; /28 mask = 255.255.255.240

Containers, Kubernetes & DevOps

How modern AI is deployed and scaled

SOFTWARE ENGINEERING VS DEVOPS

SWE Builds the software: features, logic, APIs. 'Does it work?'
DevOps Ships and runs it: CI/CD, IaC, containers, monitoring. 'Deliver at scale?'
Cousins SRE (reliability), Platform eng, DevSecOps (shift left), MLOps (models)

CONTAINERS

Container vs VM Shares host OS kernel (light) vs full guest OS (heavy)
Docker Dockerfile (recipe) -> image (template) -> container (running) -> registry (store)

KUBERNETES (ORCHESTRATOR)

Objects Pod, Node, Cluster, Control plane, Deployment, Service, Ingress, Namespace
Why for AI GPU scheduling, autoscaling inference, batch jobs, MLOps (Kubeflow/KServe)

THE NETWORKING UNDERNEATH (YOUR FOUNDATION)

Pods Each pod gets an IP; CNI wires the network; overlay (VXLAN)
Services ClusterIP (internal), NodePort, LoadBalancer; kube-proxy load-balances
Routing/DNS Ingress routes external HTTP/S; CoreDNS resolves Service names
Firewall Network Policies = least privilege for pod-to-pod traffic

THE CLOUD + DELIVERY

Managed K8s EKS (AWS), AKS (Azure), GKE; serverless containers: Fargate, Cloud Run
Pipeline CI/CD (GitHub Actions), IaC (Terraform), GitOps (Argo CD), Prometheus/Grafana

STUDY & CERTS

Path Docker -> K8s basics -> networking -> managed cluster -> CI/CD
Certs Docker DCA; CKAD/CKA; Terraform Associate; CCSP for the security seam